HDFC Bank: the path back to 2% RoA

Steering committee discussion document

What restores standalone RoA to 2.0% or more by FY29, what each move is worth, and what to start in the next 12 months

3 November 2026 | Data to Q2 FY27 | Prepared by Anklesh

Independent outside-in case study using public data only; not commissioned by, or affiliated with, HDFC Bank or Bain & Company.

HDFC Bank can be back above 2% RoA by FY29, but only by pulling six levers worth 18 bp. The share price already assumes it

Diagnosis. RoA held at 1.8-2.0% for a decade, then the merger reset the mix: NII fell from 3.83% of assets (FY23) to 3.11% (FY26). The 43 bp gap to ICICI Bank is margin (63 bp), not cost. Q2 FY27 RoA is 1.75% annualised.

Base case. RoA drifts back to 1.86% by FY29E as ₹1,93,210 cr of legacy borrowings mature within a year. Credit cost normalising to its through-the-cycle level (0.69% vs 0.48% recently) absorbs about 10 bp.

Regulation is small for the bank, not for the group. The IRDAI caps cost ₹873 cr of fees by FY29E (1.3% of other income). UPI MDR adds ₹564 cr. HDFC Life's VNB nets +₹279 cr if half the commission saving is kept.

Six levers add 17.8 bp, taking FY29E RoA to 2.04%. Four carry the plan: re-mix advances (5.3 bp), branch productivity (3.8), CASA (3.4), borrowing retirement (3.0).

Value at stake. At a 13.1% cost of equity the base case justifies ₹533 a share (sum of the parts) against ₹700 today. The price implies 17.9% RoE; the base case reaches 14.6%. Each 10 bp of RoA is worth about ₹84k cr.

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules); HDFC Bank IR26 p234; Q2 FY27 investor presentation; NSE prices 2 Nov 2026

Three decisions for the steering committee

1. Make deposits the binding constraint, not loans

Hold advances growth to the deposit franchise (credit-deposit ratio about 93%) and fund the CASA rebuild before loan share. CASA is 34.1% (FY26) vs 38% before the merger.

2. Retire ₹75,000 cr of legacy borrowings ahead of plan

Borrowings cost 7.80% vs 4.76% for deposits. ₹1,93,210 cr matures in FY27; refinancing less of it is worth 3.0 bp.

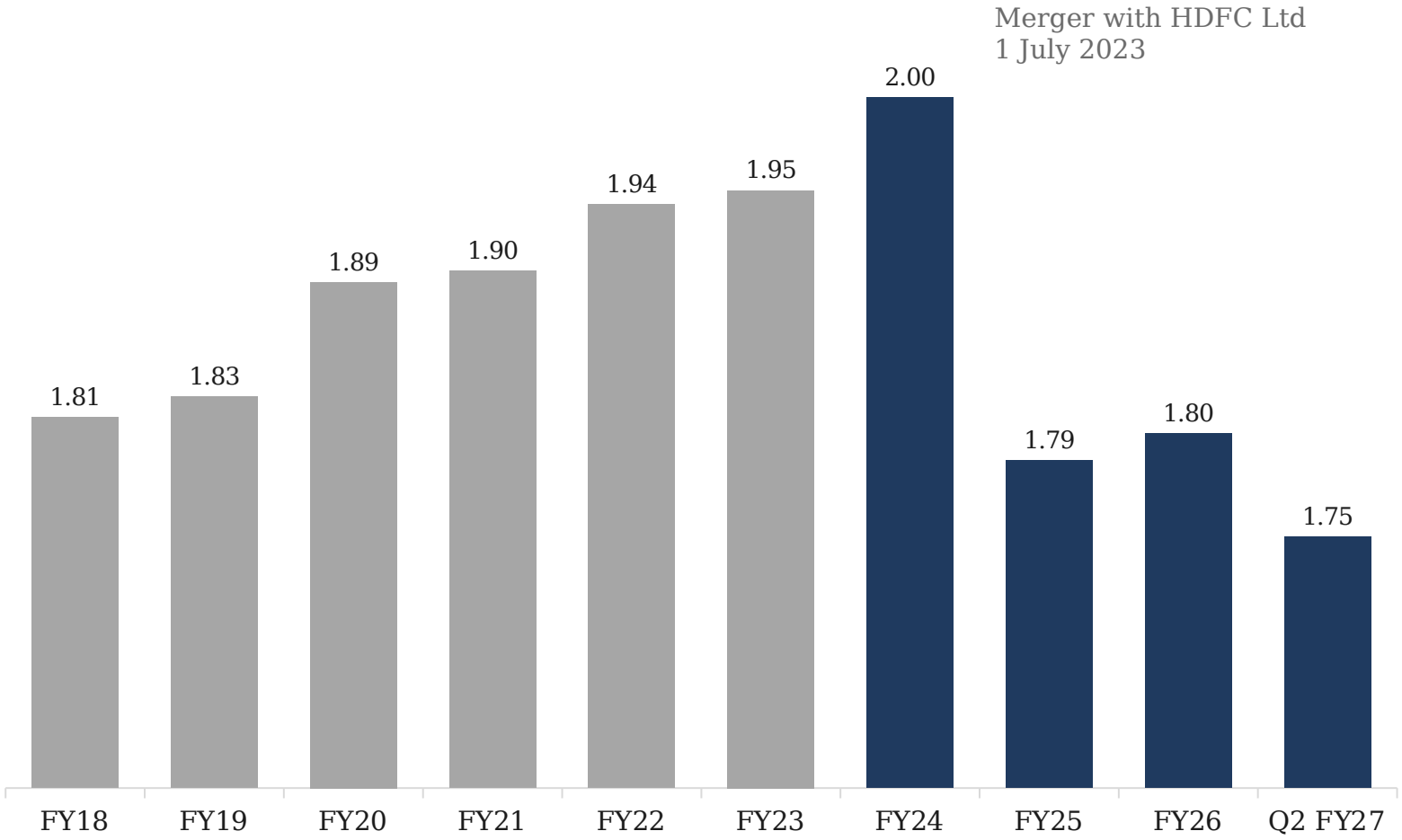
3. Re-price the HDFC Life partnership before the IRDAI caps land

Move bancassurance pay to persistency-linked commission and longer premium terms, so the group keeps the margin the caps create.

Source: Team analysis; lever values from HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules)

SITUATION

For a decade HDFC Bank delivered 1.8-2.0% RoA; since the merger it has sat at the bottom of that range



1.75%

Q2 FY27 RoA, annualised. Reported 1.78%

3.16%

Q2 FY27 NIM on total assets, down 22 bp from Q4 FY26

32%

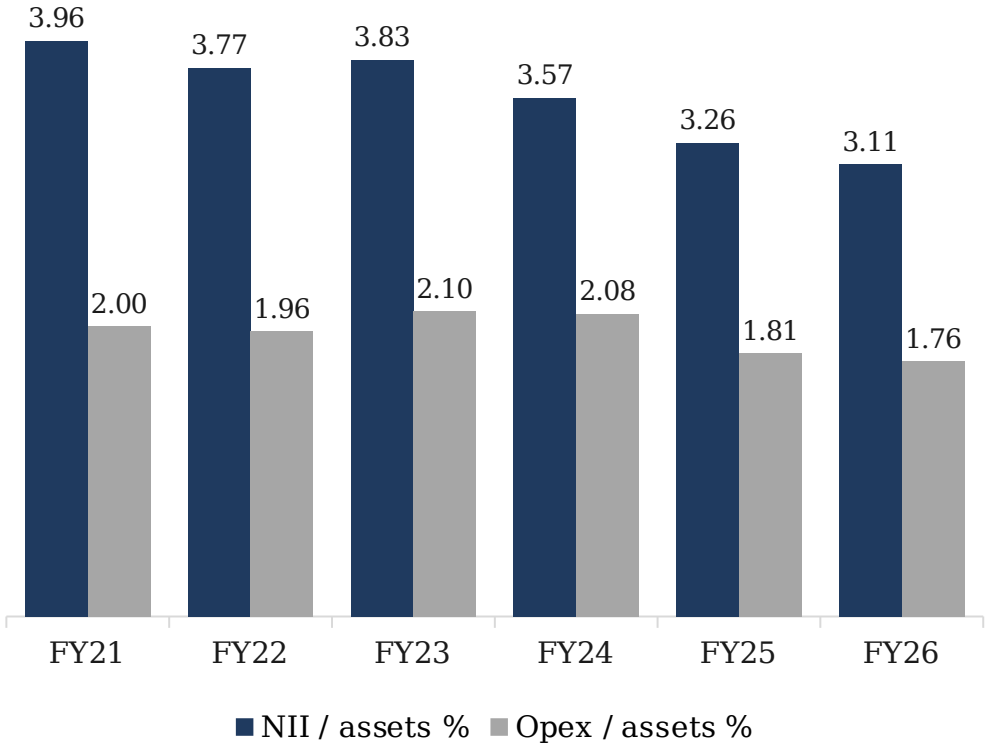
CASA ratio, Q2 FY27, vs 38% before the merger

Source: HDFC Bank IR26 p234 (ten-year highlights, standalone); Q2 FY27 standalone results via Screener, tied to the Q2 FY27 presentation. RoA = PAT / simple average total assets; Q2 FY27 annualised

The merger moved the problem from cost to margin: NII lost 72 bp of assets while opex fell 35 bp

% of average assets	FY23 (pre-merger)	FY26	RoA effect, bp
NII	3.83%	3.11%	-72
Other income	1.38%	1.51%	+13
Opex	2.10%	1.76%	+35
Provisions	0.53%	0.57%	-4
Tax	0.63%	0.50%	+14
RoA	**1.95%**	**1.80%**	**-14**

So what. Cost is already best-in-class among large private banks. The recovery has to come from funding cost and asset mix.



Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Model tab (ratios on average total assets). FY26 includes a ₹12,861 cr gain on sale of investments and ₹10,700 cr of floating and contingent provisions

COMPLICATION

The market is paying for 17.9% RoE; the base case delivers 14.6% by FY29

1.92x

Price to standalone book today

17.9%

RoE implied by that P/B at Ke 13.1% and g 8%

14.6%

Base-case RoE, FY29E

The gap is about 3 percentage points of RoE. Closing it through RoA alone needs roughly 41 bp, since leverage is fixed by the Tier 1 ratio.

Justified P/B in the base case is 1.29x on FY29E RoE, which gives ₹533 a share once listed subsidiaries are added at market value less a 20% holding-company discount.

The levers are therefore not optional. Without them the stock carries a de-rating risk; with them and the upside scenario it is fairly priced.

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Valuation tab. Cost of equity by CAPM: 10-year G-sec 6.96% (RBI, Aug 2026), beta 0.95, equity risk premium 6.5%. Price ₹700 and market capitalisation from NSE, 2 Nov 2026

APPROACH

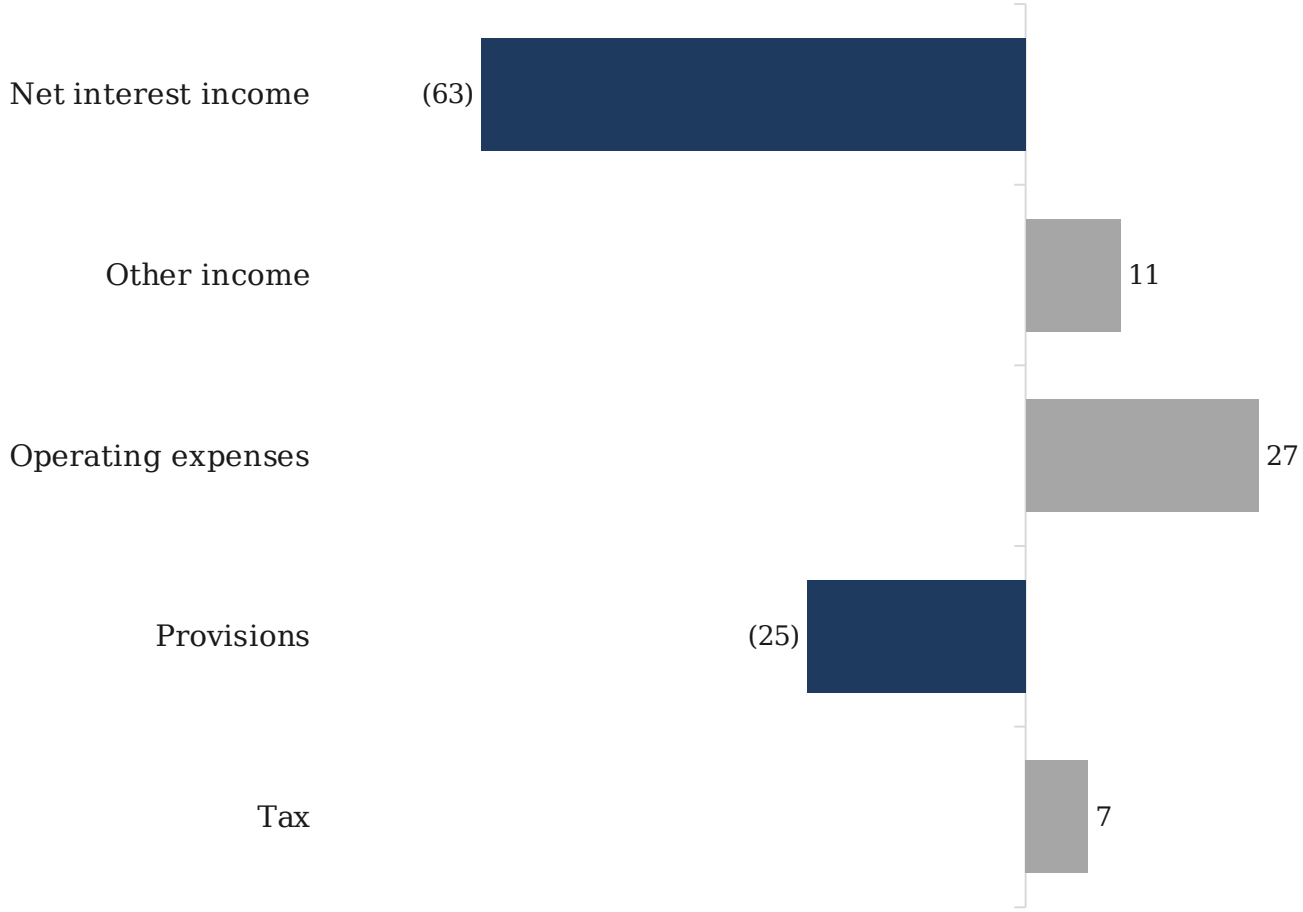
We tested five hypotheses, one per branch of the RoA tree

How does HDFC Bank get standalone RoA to 2.0% or more by FY29?

	Hypothesis	Finding
Margin	Legacy borrowings run off mechanically; CASA must be rebuilt	Supported: borrowings fall from 19.2% to 7.2% of funding
Fees	IRDAI caps hurt the group more than the bank; UPI MDR is small	Supported: bank fee loss ₹873 cr; MDR ₹564 cr
Cost	Opex/assets keeps falling as new branches mature	Partly: 1.76% to 1.69%, about 6 bp
Credit	Credit cost is below cycle; normalisation is a headwind	Supported: 21 bp gap on loans, about 10 bp of RoA
Value	The market price needs RoE well above the base case	Supported: implied RoE 17.9% vs 14.6%

Source: Team analysis

Against ICICI Bank the 43 bp RoA gap is margin plus FY26 provisions; HDFC Bank is already ahead on cost and fees



Accent bars widen the gap. NII costs 63 bp and provisions 25 bp, most of it the ₹10,700 cr floating and contingent build in FY26.

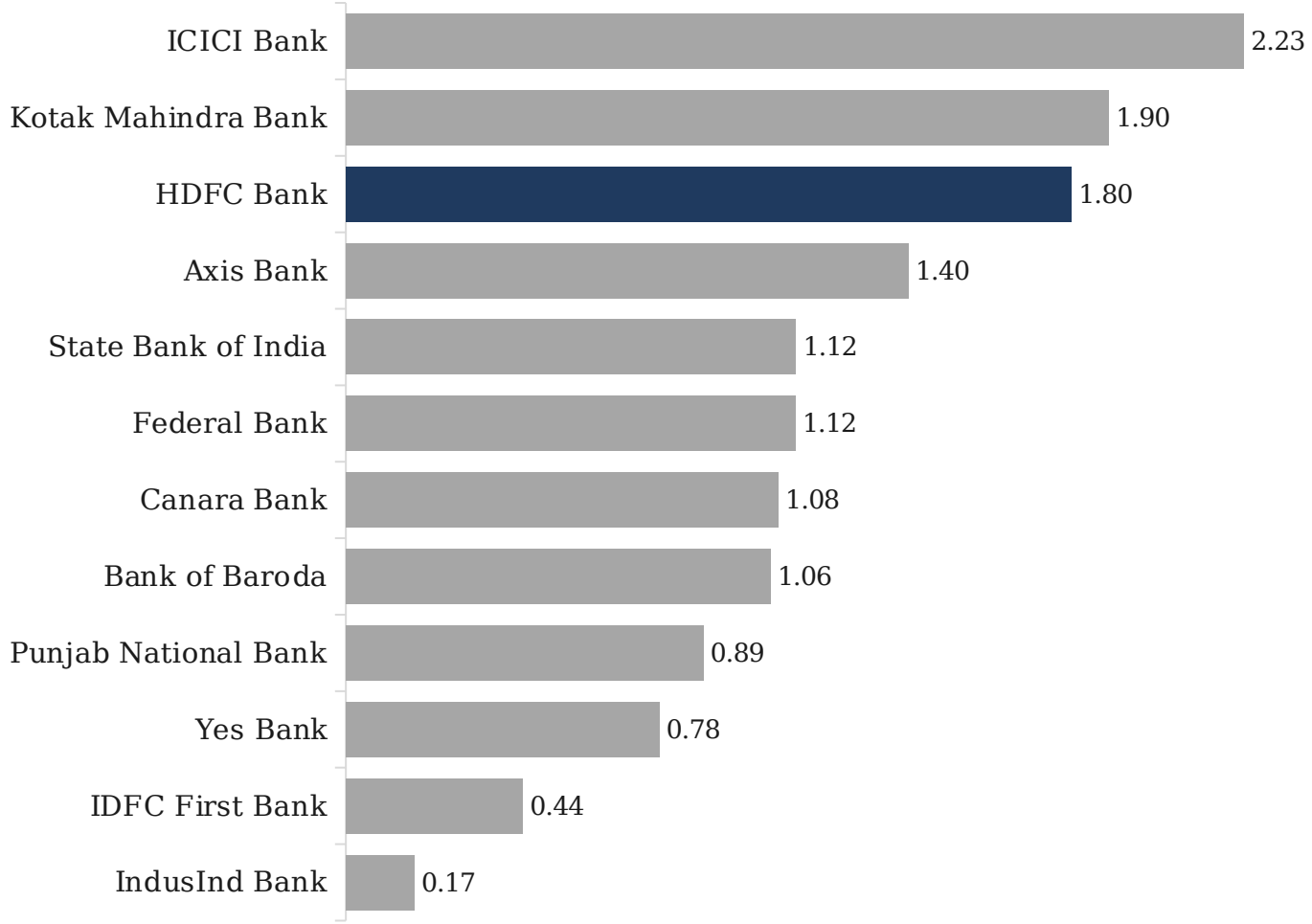
Grey bars narrow it. Better other income, opex and tax give back 45 bp.

Why. The merger brought a mortgage book at thin spreads funded by wholesale borrowings, so yield fell and cost of funds rose at the same time.

Implication. The provisions gap reverses as the one-off unwinds; the margin gap does not. The fix is funding and mix.

Source: Screener standalone financials FY26, tied to annual reports; Shapley (order-independent) attribution of the RoA gap, data/tidy/bank_shapley_FY26.csv

Among twelve listed banks HDFC Bank ranks third on RoA, behind ICICI Bank and Kotak



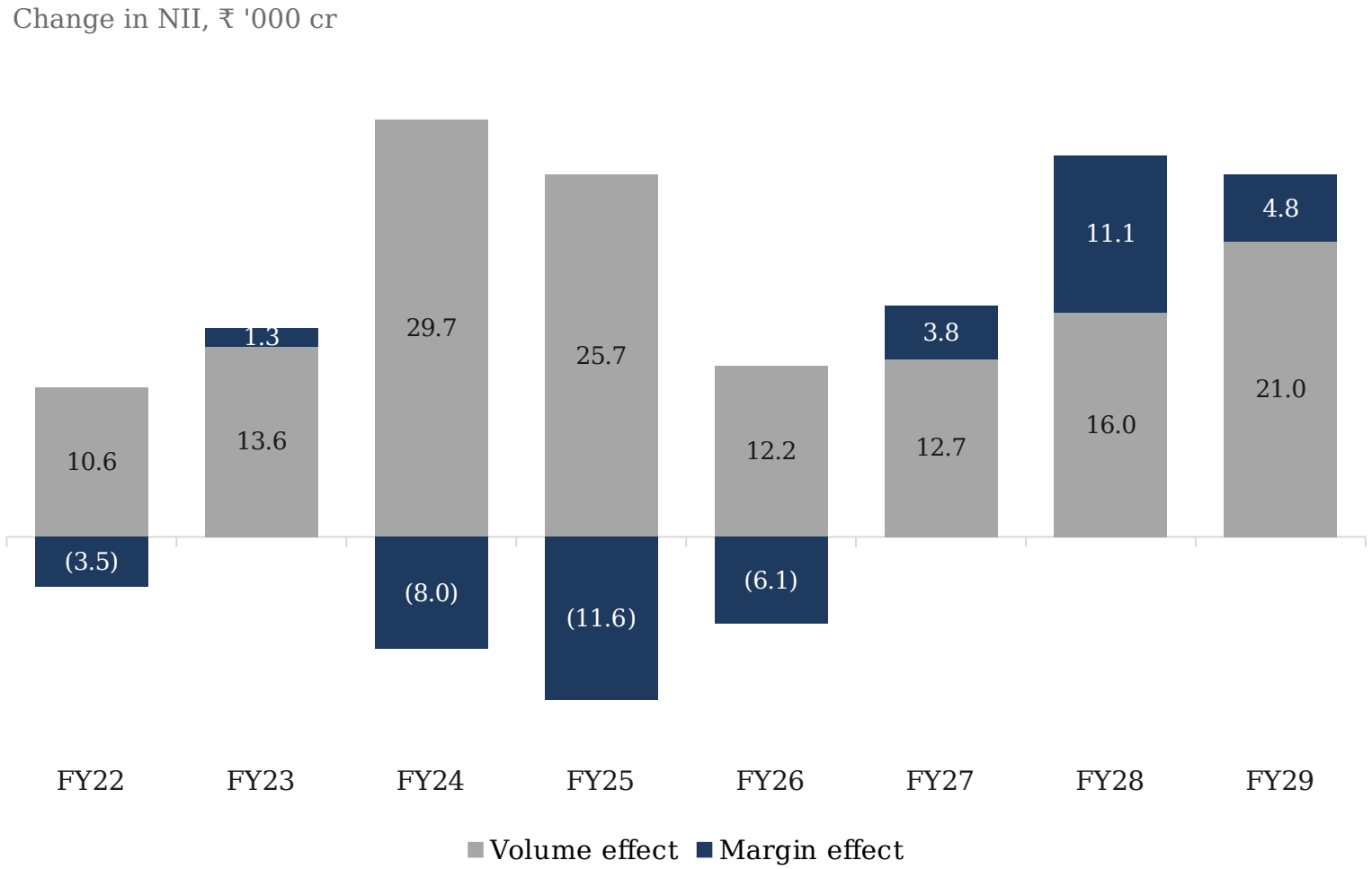
The benchmark is 2.2%, not 2.0%. ICICI Bank shows a large private bank can earn 2.2% RoA in this rate cycle.

Scale is not the excuse. HDFC Bank's assets are 1.8 times ICICI Bank's, yet it has the lower cost ratio.

PSU banks are not the peer set. SBI earns about 1.1%, but its Q2 FY27 NIM rose to 2.94%: the cycle is not hurting every bank equally.

Source: Screener standalone FY26: PAT / average total assets; Peers tab of HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules)

Margin effects took ₹25.7k cr out of NII in FY24-FY26; the model has them turning positive from FY27



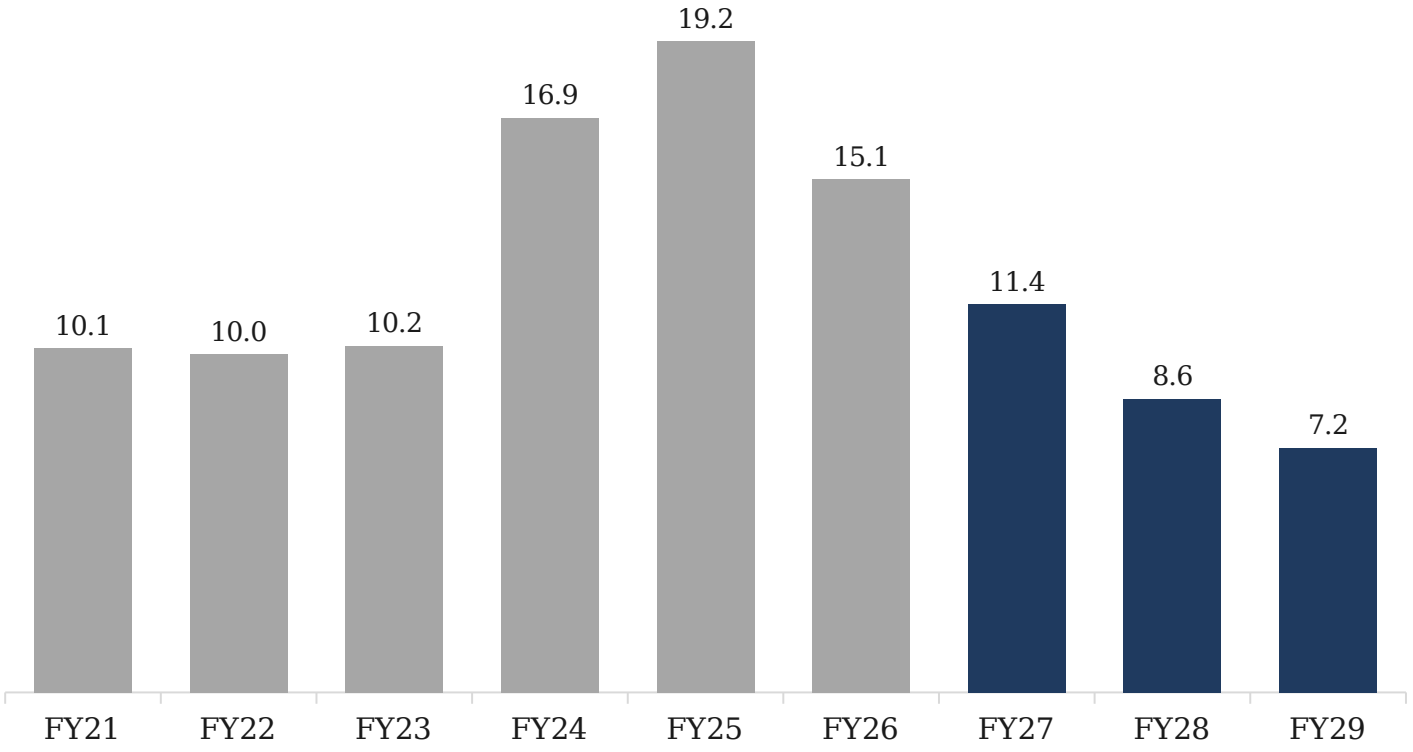
FY24-FY26: margin effects of ₹25,721 cr; growth carried NII.

FY27E-FY29E: NIM rises from 3.12% to 3.51% as borrowings run off and the October 2026 repo hike passes into loan yields.

Watch-point. The base case assumes a deposit beta of 0.35; FY26 realised beta was 0.17, which is the main risk to this walk.

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Margin tab. Volume effect = change in average interest-earning assets × prior-year NIM; margin effect = change in NIM × current assets

Legacy borrowings are the cheapest margin to win back: they fall from 19.2% of funding to 7.2% by FY29E

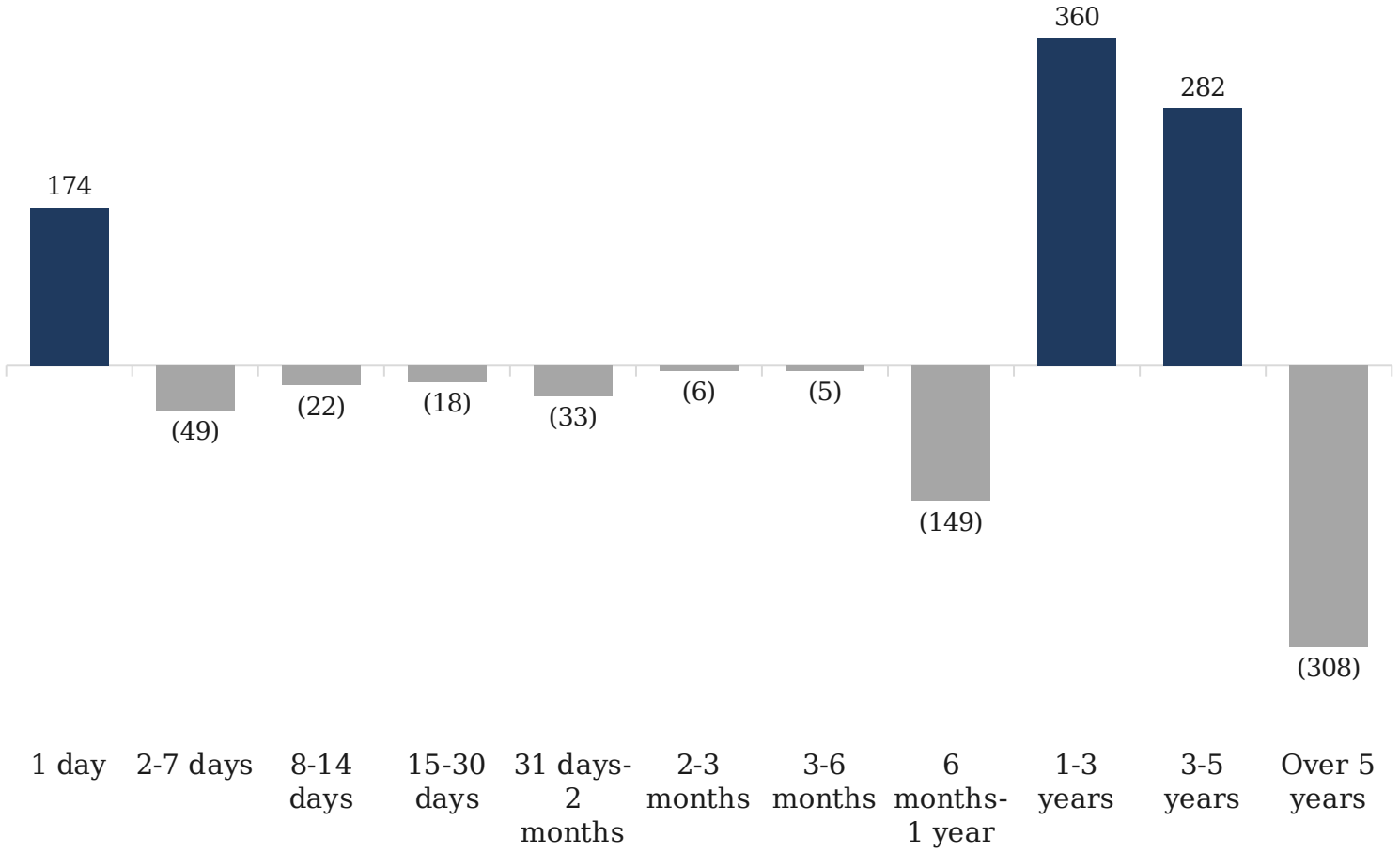


Borrowings maturing	₹ cr
Within 1 year	1,93,210
1-3 years	78,306
3-5 years	67,466
Over 5 years	1,50,413
Total, 31 Mar 2026	**4,89,395**

Each ₹1 lakh cr refinanced with deposits saves about ₹3,046 cr a year before tax.

Source: HDFC Bank IR26 p384 (maturity pattern of borrowings, 31 Mar 2026) and Schedules 4, 15; HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Margin tab

HDFC Bank is asset-sensitive: +100 bp adds ₹12,569 cr to NII, so the October repo hike helps if deposit pricing holds



12,569

Earnings at risk, +100 bp, 12 months, ₹ cr

9.8%

of FY26 NII; about 21 bp of FY27E RoA after tax

0.17

Realised deposit beta, FY25-FY26 (change in cost of deposits / change in repo)

Source: HDFC Bank IR26 p384 (maturity pattern); external-benchmark share of advances is a labelled assumption (55%, range 45-65%); RBI policy repo; HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), ALM tab

Management has already missed one margin range; the new guidance is directional, so the board should track it quarterly

Call	Speaker	What management said (paraphrase)	What happened	Verdict
Q2 FY25	MD&CEO	Margins to stay in a tight operating range (3.4-3.5%) in near and medium term	3.38 on total assets (Q4FY26); 3.16 total assets / 3.32 IEA (Q2FY27): below the stated range	Missed
Q2 FY27	CFO	NIM at the bottom; neutral-to-positive trajectory from here	Open - test at Q3FY27 results (Jan 2027)	Open

Board test for Q3 FY27 (January 2027): NIM on total assets at or above 3.16%, and cost of deposits flat or lower quarter on quarter.

Source: Earnings-call transcripts via FactIQ (links in data/inputs/concall_say_do.csv); HDFC Bank Q4 FY26 and Q2 FY27 investor presentations, p3. Quotes are paraphrased; verbatim text is at the links

The IRDAI caps cost the bank ₹873 cr of fees by FY29E; the bigger question is how the group splits HDFC Life's saving

Item	₹ cr	Basis
Bank fee lost, FY28E	437	Half the cap effect in year one
Bank fee lost, FY29E	873	1.3% of FY29E other income
HDFC Life commission above caps, all channels	1,259	Annual, on FY25 mix
HDFC Life net change in VNB	+279	Half of saving kept; 15% of banca APE lost
Value to HDFC Bank shareholders	+1,642	11.7x VNB multiple; 50.2% stake

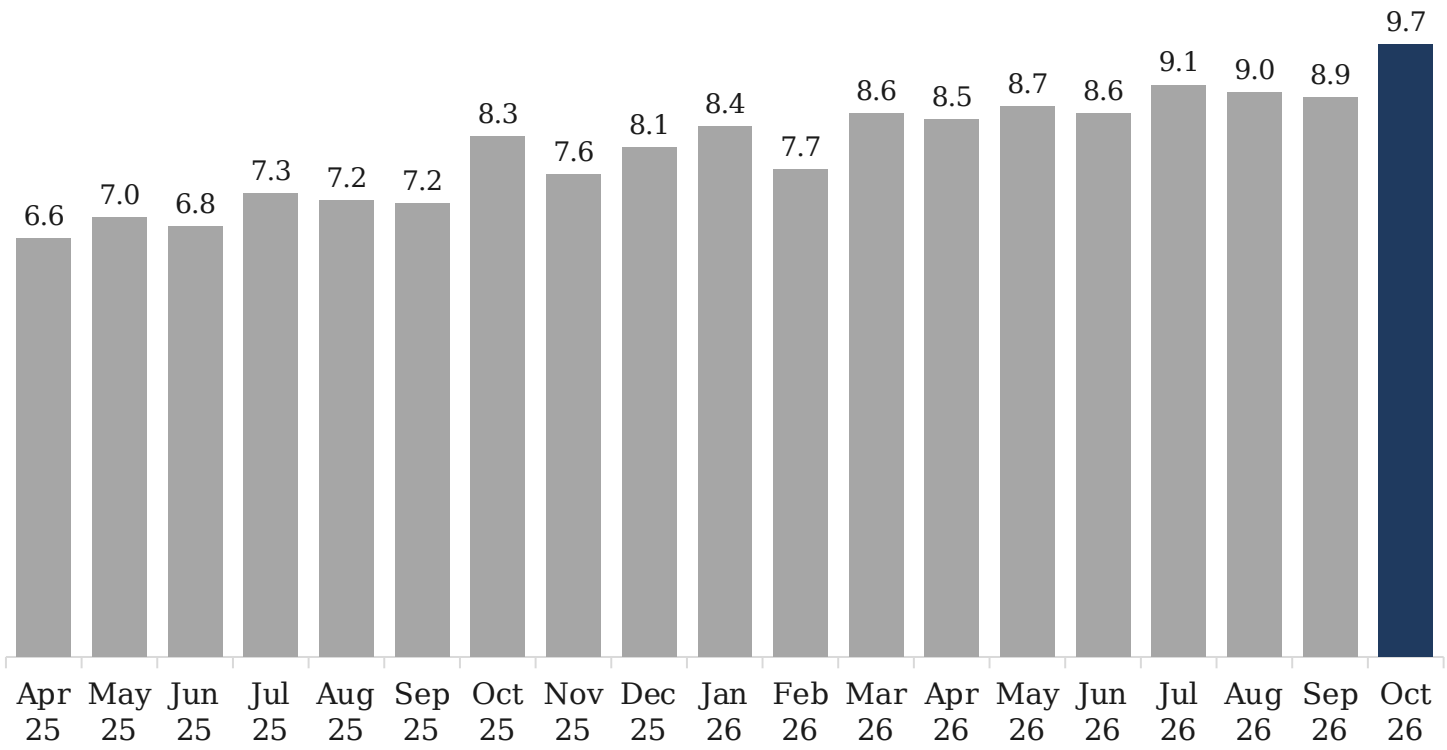
For the bank: a rounding error on other income, and it lands from FY28.

For the group: bancassurance is 58% of HDFC Life's individual APE. If the bank pulls back, volume losses can outweigh the margin gain.

Move: agree persistency-linked pay and a longer premium-term mix with HDFC Life now, before the final rules.

Source: IRDAI exposure draft on commission and expenses of management, Sept 2026 (proposal, not final); IRDAI Handbook 2024-25 T100; HDFC Life FY26 and Q2 FY27 disclosures; HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Fees tab

UPI MDR opens a ₹18.5k cr pool; at today's 6.6% payee share HDFC Bank books about ₹564 cr a year by FY29E



Merchant (P2M) UPI value per month, ₹ lakh cr. October 2026 is the record month: 25.47 bn transactions in total

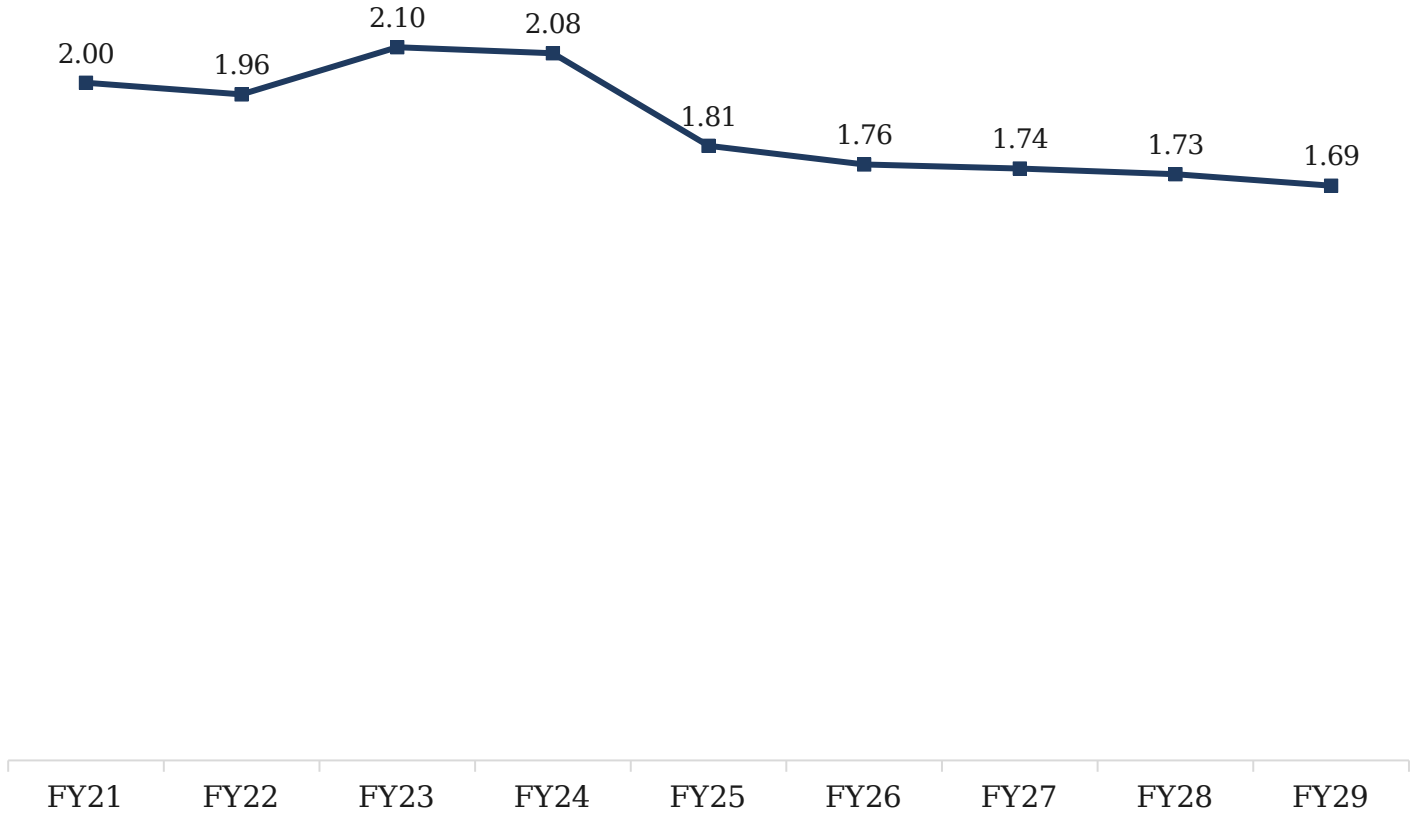
Pool. 67.7% of merchant value is above ₹2,000; after exempt categories and the ₹300 cap the pool is ₹18,541 cr a year.

HDFC Bank's take. ₹92 / 456 / 564 cr in FY27E-FY29E. Yes Bank, with about 54% of payee volume, is the structural winner.

Lever. Lifting payee share to 10% adds only 0.4 bp. UPI is a franchise play, not a RoA lever.

Source: NPCI UPI statistics via FactIQ (P2M value by ticket band, Apr 2025 to Oct 2026; payee banks Sep 2026). MDR of 0.4% above ₹2,000 per the proposal; exempt share and acquirer share are labelled assumptions. HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Fees tab

Cost-to-income falls from 38.0% to 36.4% in the base case; the branch build-out has to start paying back



₹19.7 cr

Net revenue per branch, FY26

35.9%

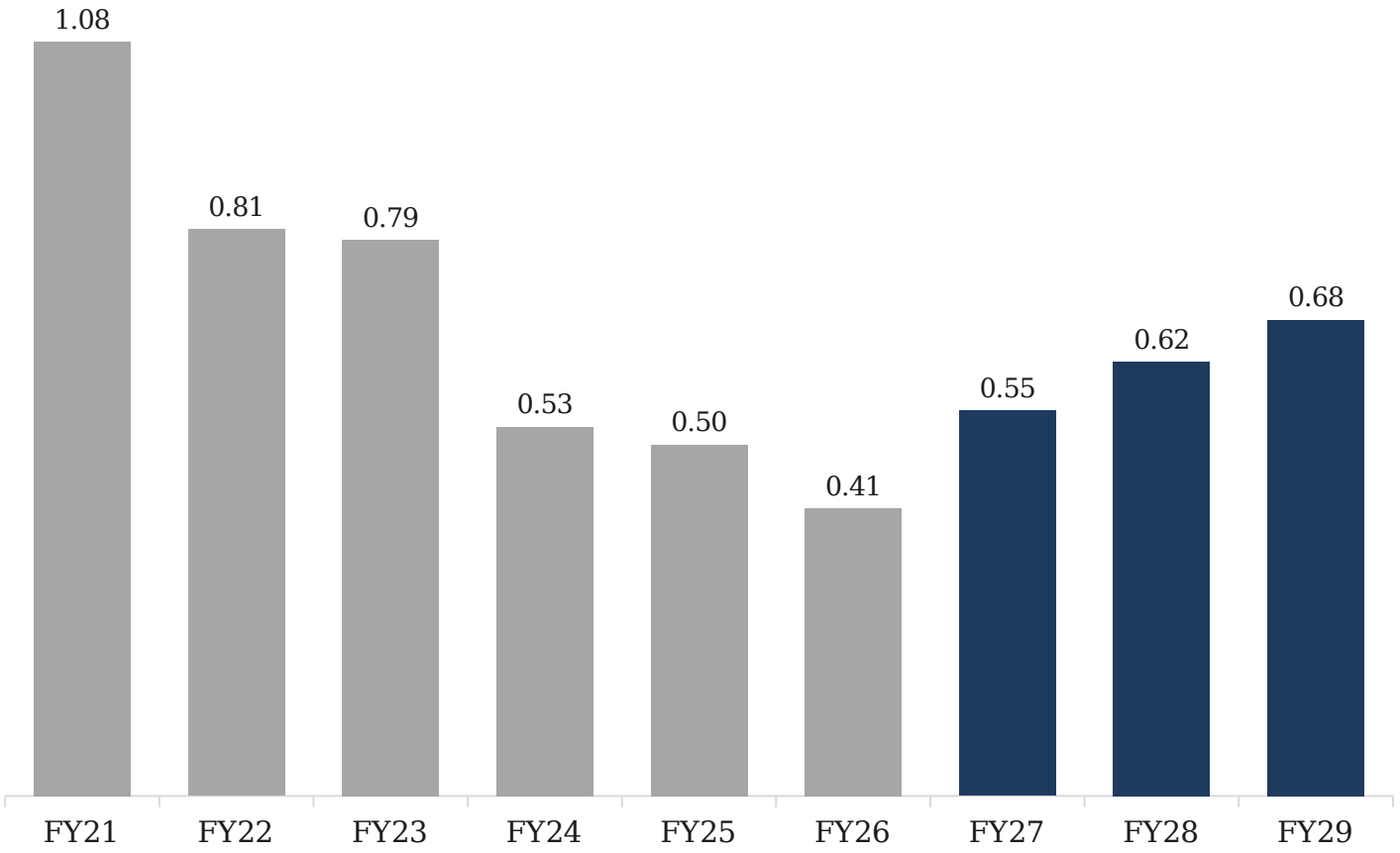
Employee cost share of opex, FY26

3.8 bp

RoA from 3% lower opex than plan by FY29E

Source: HDFC Bank IR26 Schedule 16 and p18 (9,689 branches), p420 (2,10,969 employees); HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Opex_Credit tab

Credit cost is 21 bp below its through-the-cycle level; the plan must absorb about 10 bp of RoA as it normalises



0.69%

Through-the-cycle credit cost, FY21-FY26

0.48%

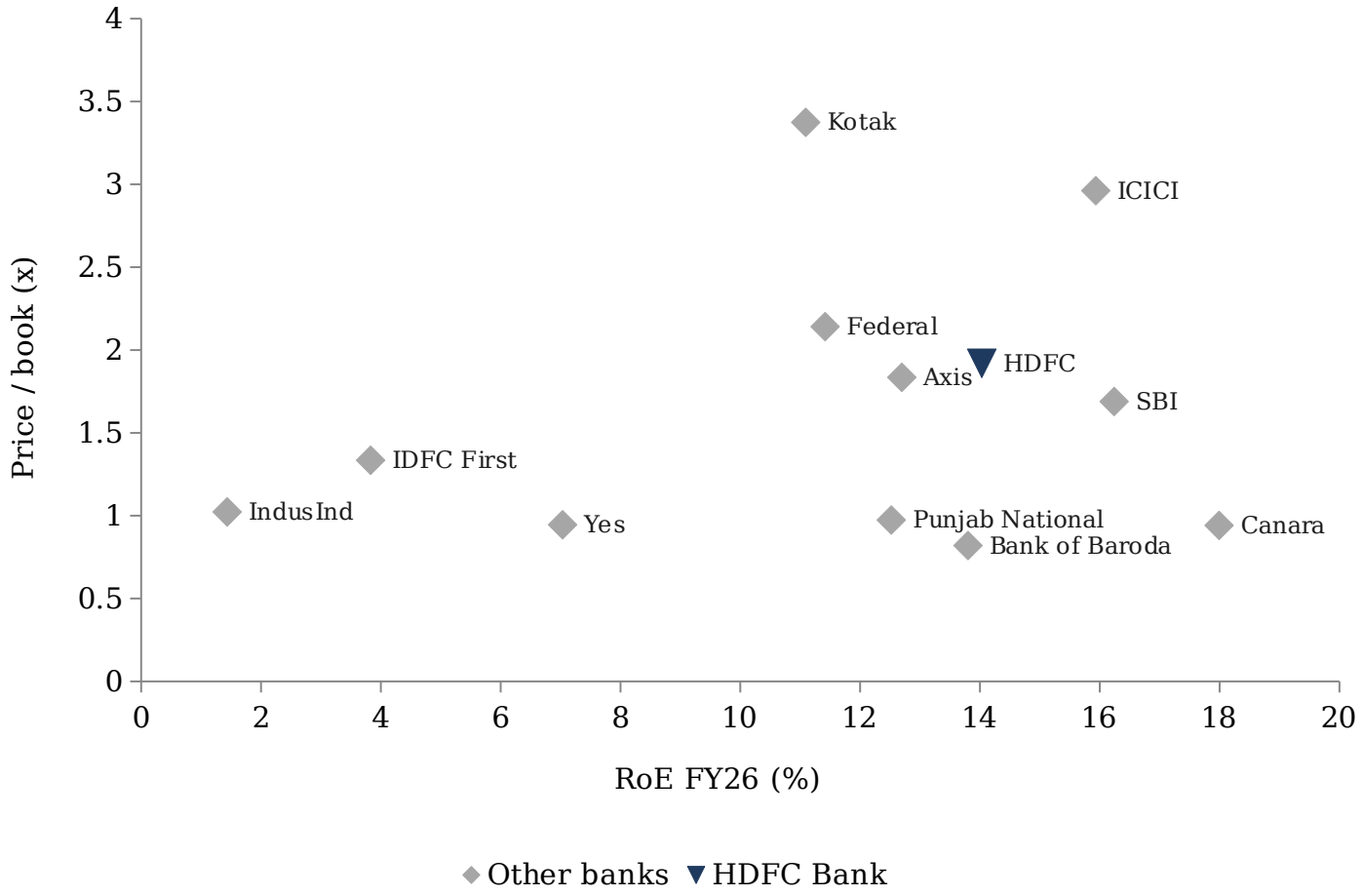
Recent average, FY24-FY26

0.85%

Downside case FY29E; RoA falls to 1.32%

Source: HDFC Bank IR26 p234 (loan loss provisions); HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Opex_Credit and Inputs tabs. FY27E-FY29E is the base-case path

Across peers P/B barely tracks RoE, so we value the levers on justified P/B rather than the regression



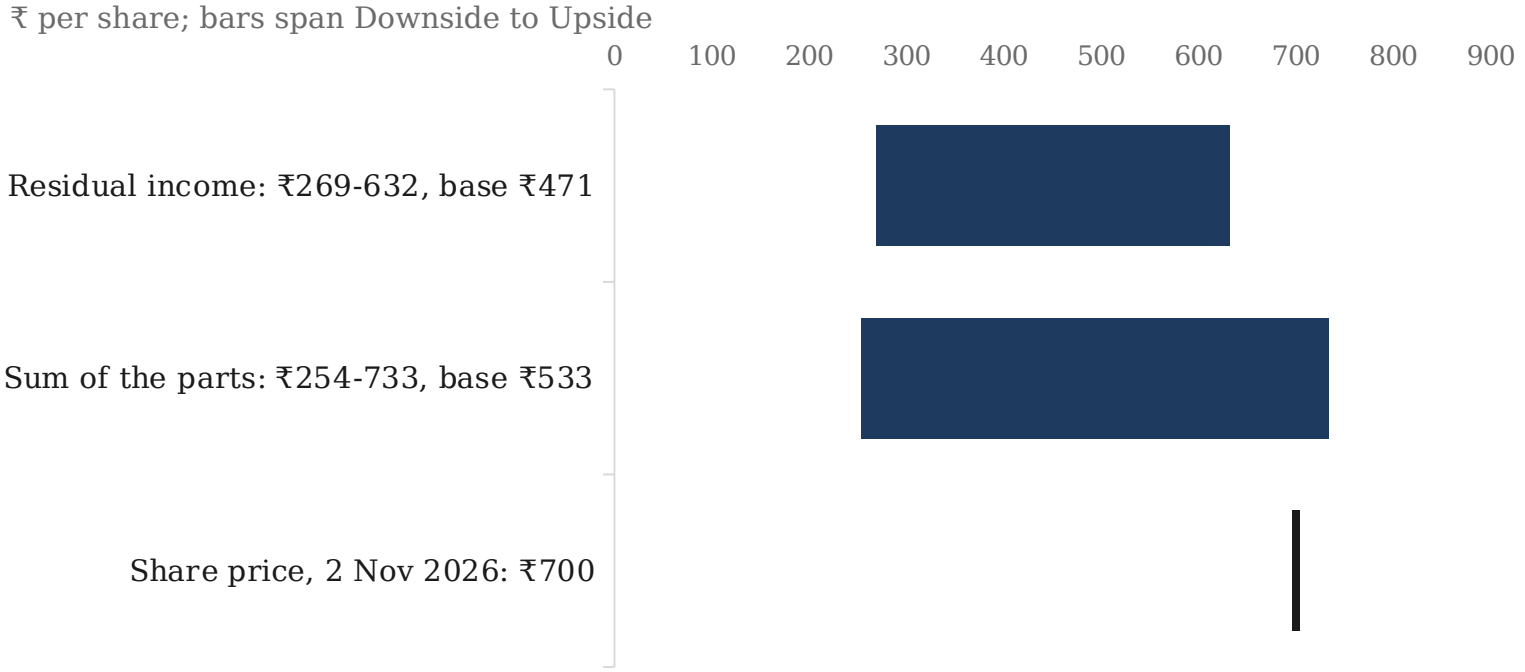
R-squared 0.001 on the five clean private banks. Kotak and ICICI Bank trade on franchise and growth, not on current RoE.

Method. Justified P/B = (RoE − g) / (Ke − g) with Ke 13.1% and g 8%. A 10 bp rise in sustainable RoA is worth ₹84,212 cr of equity value.

Cross-check. Residual income on FY27E-FY29E with a fading terminal value gives a similar answer.

Source: Screener market capitalisation and book value, 2-3 Nov 2026; FY26 standalone RoE; Peers tab of HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules)

Only the upside case supports today's ₹700 share price; the base case is worth ₹471-533



Upside needs: repo held at 6.25%, CASA back to 36%, 80% of maturing borrowings retired and credit cost at 0.60% by FY29E.

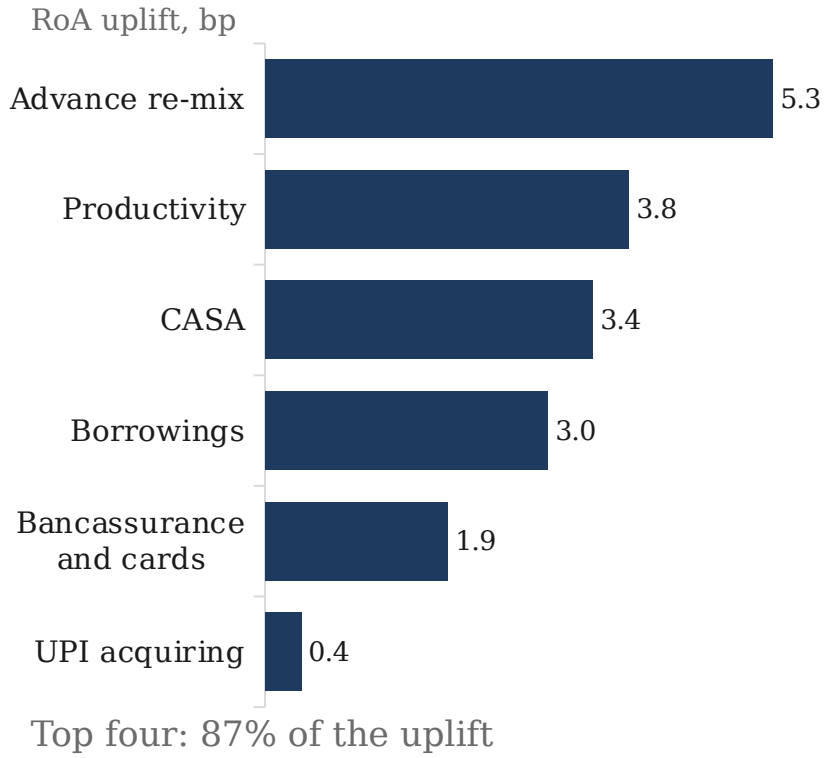
Downside: faster credit normalisation (0.85%) and sticky deposit costs take RoA to 1.32%. The levers cannot close that gap.

Scenario	RoA FY29E	RoE FY29E	Residual income / share	Sum of the parts / share	vs ₹700
Downside	1.32%	10.6%	₹269	₹254	-63.8%
Base	**1.86%**	**14.6%**	**₹471**	**₹533**	**-23.9%**
Upside	2.25%	17.5%	₹632	₹733	4.8%

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Valuation tab, recalculated per scenario. Bars span Downside to Upside; the base case is labelled

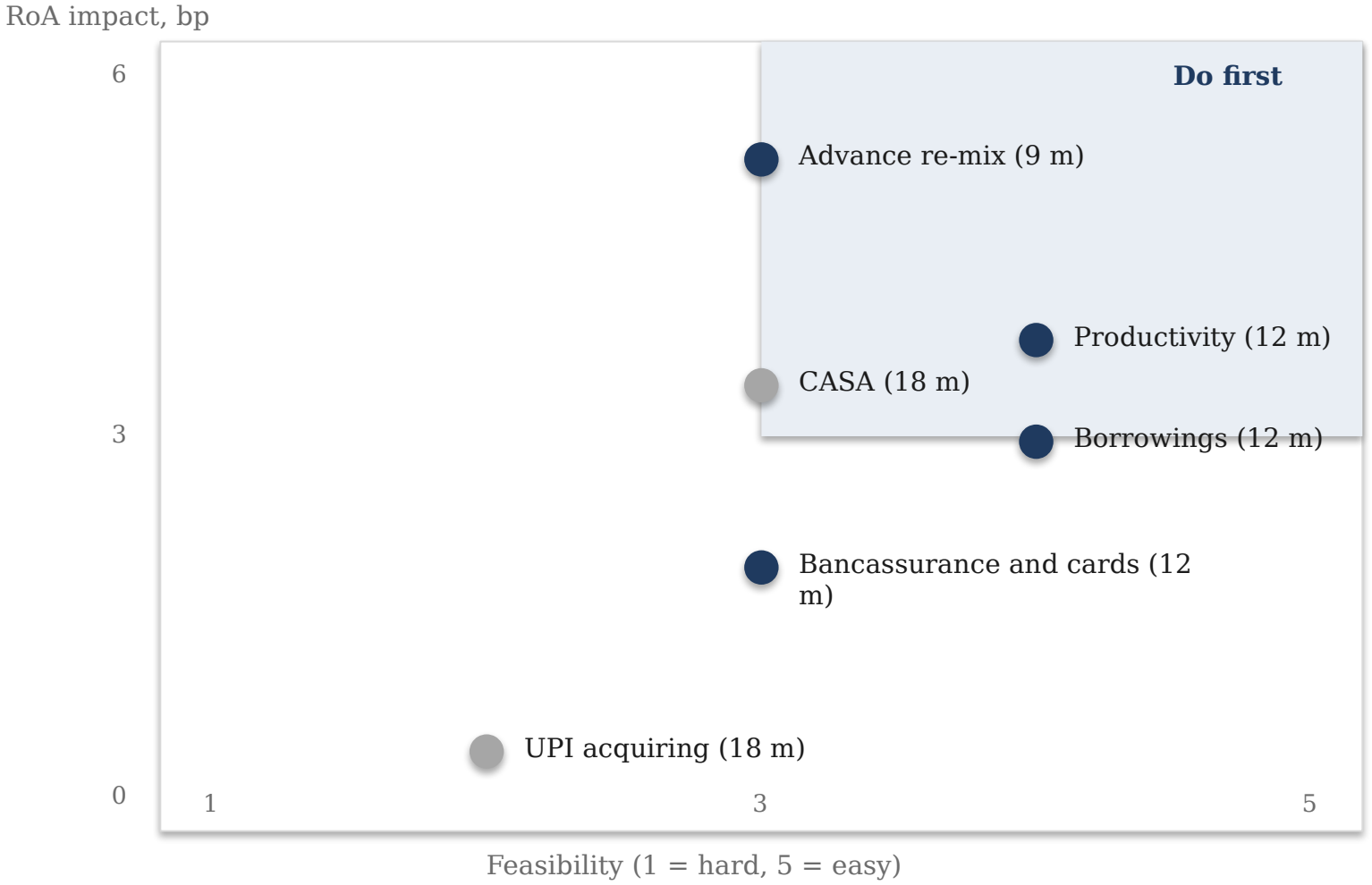
Six levers add 17.8 bp of RoA and ₹150k cr of equity value; four of them do 85% of the work

Lever	RoA, bp	PAT, ₹ cr	Equity value, ₹ cr	Per share
Re-mix advances to higher-yield retail and SME	5.3	3,030	44,673	₹29
Branch and operations productivity	3.8	2,171	32,015	₹21
Rebuild CASA	3.4	1,956	28,839	₹19
Retire legacy borrowings faster	3.0	1,688	24,889	₹16
Protect bancassurance and grow card fees	1.9	1,090	16,075	₹10
Win UPI merchant acquiring share	0.4	215	3,174	₹2
All levers	**17.8**	**10,150**	**1,49,666**	**₹97**



Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Levers tab. Each lever is a driver change on top of the base case in FY29E; sizes, feasibility and timing are judgements for discussion

Start with borrowings and productivity: both are high-feasibility and pay back within a year



Wave 1 (0-12 months): retire borrowings, branch and operations productivity, advance re-mix.

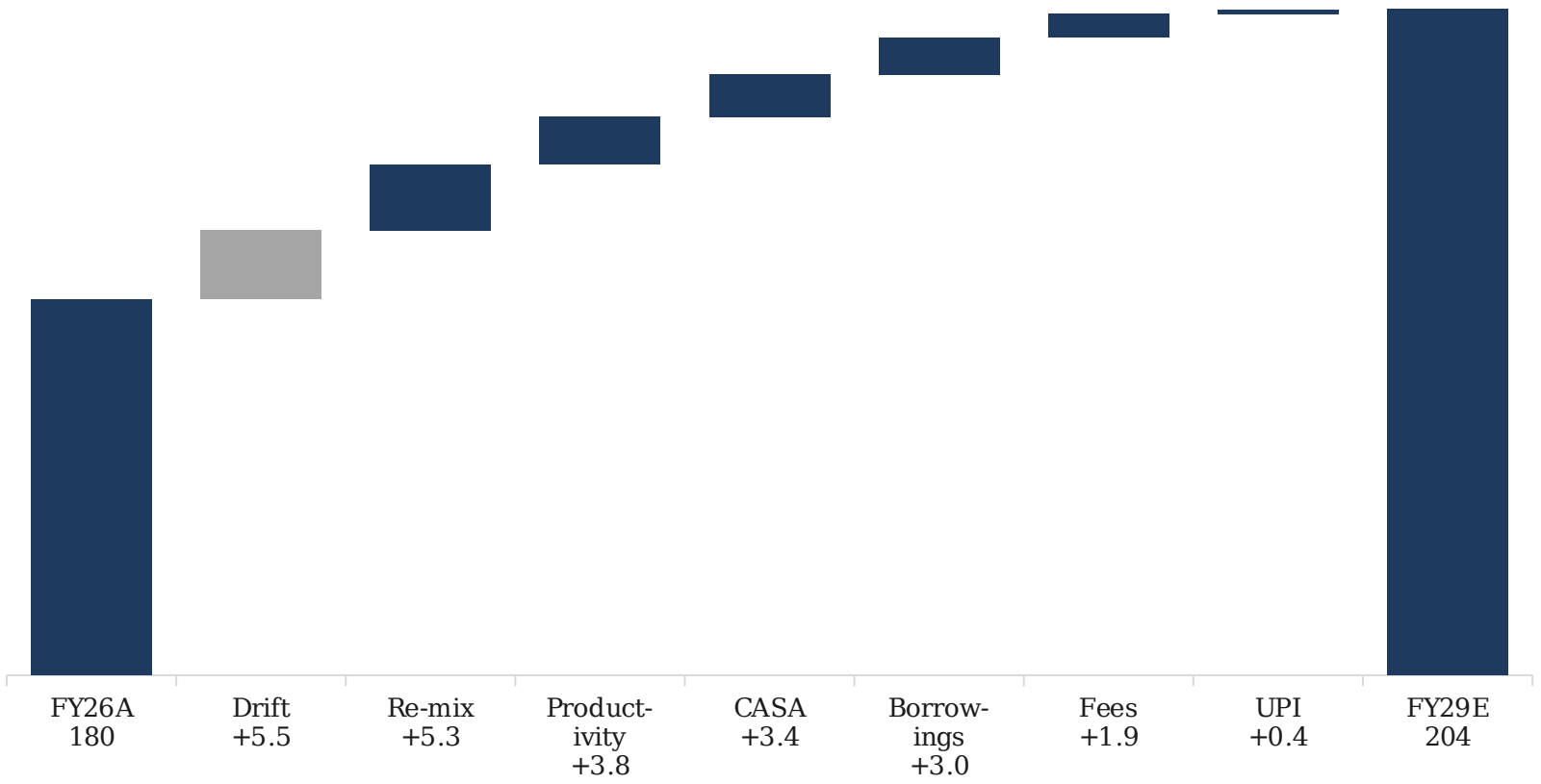
Wave 2 (12-18 months): CASA rebuild, which is the largest structural lever but slowest; UPI acquiring as a franchise option.

Blue dots pay back within 12 months; grey take longer.

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Levers tab (feasibility 1-5 and months to impact are team judgements)

FY26 to FY29E: the base case adds 5.5 bp, the levers 17.8 bp, for 2.04% RoA against a 2.00% target

RoA, bp; axis starts at 150 bp



bp of RoA, FY29E

Scenario	Before levers	Levers	With levers	vs 200
Downside	132	18.0	150	-50
Base	**186**	**17.8**	**204**	**+4**
Upside	225	17.7	243	+43

Downside is the real exposure: the levers add 18 bp, but RoA still ends 50 bp short of target.

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Levers tab (bridge). Core RoA excludes the FY26 one-offs

A 12-month plan: three moves start now, with board checkpoints at each quarterly result

Move	0-3 months	3-12 months	Owner
Funding: retire borrowings	Approve refinancing plan for ₹1.93 lakh cr maturing in FY27; set deposit-led target	Retire ₹75,000 cr above plan; borrowings below 9% of funding	Treasurer
Funding: rebuild CASA	Salary and SME current-account squads in the top 500 branches; re-price term deposits by tenor	CASA +1.5 pp vs plan; cost of deposits flat while repo rises	Head, Branch Banking
Assets: re-mix advances	Shift new mortgage origination to pricing floors; grow SME and secured retail within risk appetite	Yield on advances +10 bp vs plan	CRO and Head, Retail Assets
Cost: productivity	Branch-vintage review of FY22-FY25 openings; freeze net hiring outside sales	Opex 3% below plan; cost-to-income under 38%	COO
Fees: bancassurance	Open renegotiation with HDFC Life on persistency-linked pay; model final IRDAI rules	New commission grid live before caps take effect	Head, Third-party Products
Governance	RoA cockpit to the board monthly; lever owners named	Quarterly lever review at each result	CFO

Source: Team analysis

Seven KPIs tell the board whether the plan is on track

KPI	FY26A	FY29E base	FY29E with levers	Cadence
RoA (core)	**1.80%**	**1.86%**	**2.04%**	**Quarterly**
NIM on interest-earning assets	3.12%	3.51%	–	Quarterly
CASA ratio	34.1%	33.5%	35.0%	Monthly
Borrowings / funding	15.1%	7.2%	−₹75,000 cr	Monthly
Opex / average assets	1.76%	1.69%	1.64%	Quarterly
Credit cost	0.41%	0.68%	–	Quarterly
RoE	14.0%	14.6%	16.0%	Quarterly

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules); FY26 from HDFC Bank IR26. With-lever values apply lever sizes to the base case

Credit normalisation and deposit pricing are the two risks that could break the plan

Risk	Early warning	RoA impact	Mitigation
Credit cost normalises faster	Slippages above 1.5% annualised	54 bp (downside case, all drivers)	Tighten unsecured origination; hold floating provisions
Deposit costs stay sticky	Cost of deposits rises faster than repo	Deposit beta 0.35 to 0.50 removes most of the NIM gain	CASA squads; tenor pricing
Rate cuts in FY28	Repo below 5.25%	About 21 bp per 100 bp (asset-sensitive)	Raise fixed-rate share of new loans
Final IRDAI caps stricter	Final rules at or below draft limits	Up to ₹1,310 cr of fees (1.5x severity)	Persistency-linked pay agreed early
Execution of CASA rebuild	CASA flat for two quarters	3.4 bp lever at risk	Named owner; monthly tracking

Source: HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules), Inputs, ALM and Opex_Credit tabs; IRDAI exposure draft Sept 2026

Appendix

Method and model | Scenario drivers | Sector modules: IRDAI caps and SBI Q2 FY27 | Sources

Method: a driver-based bank model, valued on justified P/B, residual income and sum of the parts

Spreading. FY20-FY26 standalone P&L and balance sheet from HDFC Bank IR26 schedules and ten-year highlights, tied to Screener; one-offs isolated.

Projection. Interest on average balances; yields and costs move with repo through betas; investments and liquidity balance the sheet; in-year retained profit is held as cash and deployed the next year, so the model has no circularity.

Margin and ALM. Rate/volume walk of NII; repricing gap from the IR26 maturity pattern; earnings at risk for +100 bp; legacy borrowing run-off from the maturity schedule.

Fees. IRDAI cap model (Monte Carlo, 10,000 draws) for commission above the proposed caps; UPI MDR pool from NPCI ticket-band data and payee-bank shares.

Valuation. CAPM cost of equity; justified P/B on FY29E RoE; residual income FY27E-FY29E with terminal value; listed subsidiaries at market value less 20%.

Controls. Nine model checks roll up to a master check on the Cover; a lint script fails the build on non-Garamond fonts, italics, merged cells, hard-codes inside formulas, broken row patterns and error values.

Source: Team analysis; model is models/HDFC_RoA_Path_FY29.xlsx

Scenario drivers, FY29E

Driver (FY29E)	Base	Upside	Downside
Policy repo, fiscal-year average	5.75%	6.25%	5.00%
Advances growth	14%	16%	12%
Deposit growth	14%	16%	12%
CASA ratio, year end	33.5%	36.0%	32.0%
Maturing borrowings retired	60%	80%	30%
Deposit cost beta	0.35	0.30	0.25
Core fee growth	12%	14%	9%
Opex growth	10%	9%	12%
Credit cost	0.68%	0.60%	0.85%
IRDAI severity on bank fees	1.0x	0.5x	1.5x

Read the downside deposit beta with care: a lower beta in a falling-rate path means deposit costs lag the cut, which is adverse for an asset-sensitive bank.

Source: Inputs tab of HDFC_RoA_Path_FY29.xlsx (driver model built on HDFC Bank IR26 schedules); repo path from RBI Survey of Professional Forecasters, round 102 (Sept 2026)

Sector view: commission above the proposed caps is about ₹10.8k cr a year across life insurers

₹10.8k cr

Median annual first-year commission above the caps

₹8.6-13.2k cr

P10-P90 range

38.8% vs 12.1%

Private life commission growth vs premium growth, FY25

Who carries it. Bank-promoted insurers: the sum is about 74% of SBI Life's FY25 profit and 70% of HDFC Life's, against about 6% of LIC's.

Status. A September 2026 proposal, not final. The premium-paying-term mix is an assumption, stress-tested by the Monte Carlo.

Source: IRDAI Handbook 2024-25; IRDAI exposure draft, Sept 2026 (proposal); Monte Carlo of premium-paying-term mix, 10,000 draws, seed 20261102 (data/tidy/ins_mc_summary.csv)

Not every bank's margin fell in Q2 FY27: SBI's NIM rose to 2.94%, which corrects our earlier sector read

Bank	NIM Q4 FY26	NIM Q2 FY27	Change, bp	Basis
HDFC Bank	3.38%	3.16%	-22	total assets
ICICI Bank	4.32%	4.18%	-14	interest-earning assets
Axis Bank	3.62%	3.36%	-26	interest-earning assets (3.47 ex FCNR/SNRB)
Kotak Mahindra Bank	4.67%	4.31%	-36	interest-earning assets
State Bank of India	2.81%	2.94%	+13	whole bank (SBI Q2 FY27 press release)

SBI Q2 FY27: NII ₹49,143 cr, operating profit ₹34,226 cr, PAT ₹23,038 cr, CASA 38.32%, credit cost 0.22%.

Read-across. SBI's stronger CASA base cushioned deposit repricing; it is the clearest evidence that funding mix, not the rate cycle alone, drives margin.

Source: State Bank of India press release, Q2 FY27 results, 3 Nov 2026; HDFC Bank, ICICI Bank, Axis Bank, Kotak Q2 FY27 investor presentations (data/inputs/nim_trajectory.csv)

Sources

S1 HDFC Bank Integrated Annual Report 2025-26: Schedules 3, 4, 8, 9, 13-16 (pp359-366), maturity pattern (p384), ten-year highlights (p234), stakes (p52).

S2 HDFC Bank, ICICI Bank, Axis Bank, Kotak investor presentations Q4 FY26 and Q2 FY27; SBI Q2 FY27 press release.

S3 Screener.in standalone financials and valuation for twelve banks and three listed HDFC subsidiaries (secondary, tied to filings).

S4 RBI: policy repo, 10-year G-sec, Survey of Professional Forecasters round 102 (via FactIQ).

S5 NPCI UPI statistics: P2M value by ticket band, payee banks (via FactIQ); monthly totals from NPCI.

S6 IRDAI Handbook 2024-25 and the September 2026 exposure draft on commission and expenses of management.

S7 HDFC Life FY26 and Q2 FY27 public disclosures: VNB, APE, embedded value, channel mix.

S8 Earnings-call transcripts (FactIQ) for the say-versus-do tracker.

Framing. Independent outside-in case study using public data only; not commissioned by, or affiliated with, HDFC Bank or Bain & Company.

Source: data/raw/manifest.csv holds a sha256 hash of every raw file